Sherman Lim

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RESEARCH INTERESTS

Memory & storage systems, LLM inference systems, hardware–software co-design.

EDUCATION

Carnegie Mellon University — Pittsburgh, PA
PhD in Computer Science — Aug 2025 – Present

- Advisor: Prof. George Amvrosiadis
- Expected graduation: May 2030

National University of Singapore — Singapore
Bachelor of Computing (Computer Science) with Highest Distinction — Aug 2019 – Jan 2023

- GPA: 4.88 out of 5.00
- Thesis: Congestion Control Speciation in QUIC
- Advisor: Prof. Ben Leong

PUBLICATIONS

Sidharth Sankhe, Felix Zhang, Umayrah Chonee, **Sherman Lim**, Jiasheng Hu, Jialin Li, and Qizhen Zhang. *"PD3: Prefetching Data with DPUs for Disaggregated Memory"*. In Proceedings of the 23rd USENIX Symposium on Networked Systems Design and Implementation (NSDI '26). May 2026.

Ayush Mishra, **Sherman Lim**, and Ben Leong. *"Understanding Speciation in QUIC Congestion Control"*. In Proceedings of the 22nd ACM Internet Measurement Conference (IMC '22). October 2022.

RESEARCH EXPERIENCE

Carnegie Mellon University — Pittsburgh, PA
Graduate Research Assistant, Parallel Data Lab — Aug 2025 – Present
Advisor: Prof. George Amvrosiadis

- Conceived and currently lead a project designing memory management techniques for language model serving with emerging High Bandwidth Flash (HBF) memory.
- Developing a vLLM prototype for GPUs with emulated HBF memory.

National University of Singapore — Singapore
Research Assistant — Apr 2025 – Jul 2025
Advisors: Prof. Jialin Li and Prof. Qizhen Zhang

- Designed a high-throughput component that detects disaggregated-memory cache misses on resource-constrained NVIDIA BlueField DPUs, enabling prefetching in PD3 (NSDI '26).

National University of Singapore — Singapore
Research Assistant — Jan 2021 – Nov 2022
Advisor: Prof. Ben Leong

- Built QUICbench, a tool for benchmarking congestion control algorithms (CCAs) across QUIC stacks, and used it to show that deployed QUIC CCAs diverge greatly from their TCP namesakes (IMC '22).

INDUSTRY EXPERIENCE

Jump Trading — Singapore
Software Engineer — Jan 2023 – Feb 2025

- Led development of latency-critical C++ systems for high-frequency trading, spanning market risk management, market data ingestion, order submission, FPGA interfacing, and embedded platforms.
- Designed a low-latency inter-process communication library tailored to a specific CPU microarchitecture.
- Mentored a summer intern, scoping and supervising a performance benchmarking project.

Jump Trading Singapore
Software Engineer Intern May 2022 – Jul 2022

- Designed and implemented a C++ framework for live-testing order submissions.

ByteDance Singapore
Software Engineer Intern May 2021 – Jul 2021

- Developed a Java database driver library for a data warehouse.

SKILLS C++, C, Python, Java, x86-64 assembly, vLLM, Linux

TEACHING

CS3210 Parallel Computing, NUS	Fall 2022
CS2106 Introduction to Operating Systems, NUS	Spring 2022
CS2040S Data Structures & Algorithms, NUS	Spring 2021
CS2040S Data Structures & Algorithms, NUS	Fall 2020

AWARDS

NUS Merit Scholarship 2019 – 2023
Full-ride, merit-based scholarship for undergraduate study at NUS.

Outstanding Undergraduate Researcher Prize (Individual), NUS 2022
Annual award that recognizes top undergraduate researchers at NUS.